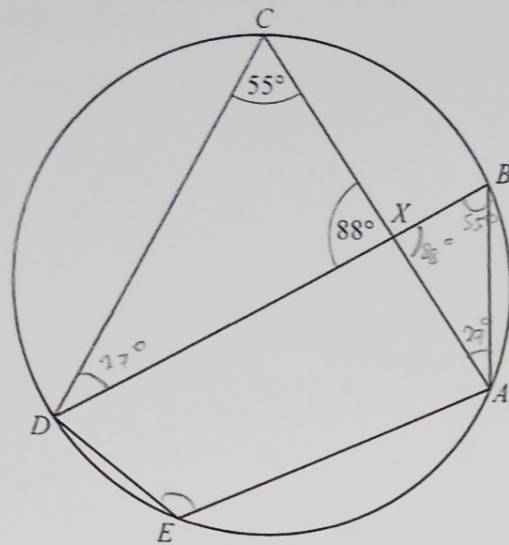




NOT TO
SCALE

A, B, C, D and E lie on the circle.
AC and BD intersect at X.
Angle $ACD = 55^\circ$ and angle $CXD = 88^\circ$.

(a) Complete the statements, giving a geometrical reason in each part.

Angle $CDB = 27^\circ$ because The sum of interior angles in a triangle add up to 180°

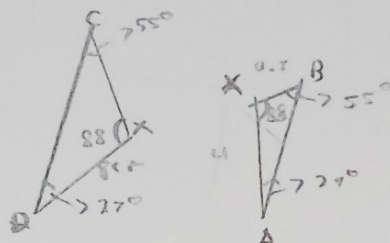
Angle $ABD = 55^\circ$ because Angles subtended from the same arc

Angle $AED = 22.5^\circ$ because opposite angles of a cyclic quadrilateral add up to 180°

[6]

- (b) Triangle CXD is mathematically similar to triangle BXA .
 $DX = 8.0$ cm, $BX = 2.7$ cm and $AX = 4.0$ cm.

- (i) Work out the length of CX .



$$\frac{8}{4} = \frac{CX}{2.7}$$

$$2 \times 2.7 = CX$$

$$5.4 \text{ cm}$$

$$\frac{DX}{AX} = \frac{CX}{BX}$$

$$CX = 5.4 \text{ cm [2]}$$

- (ii) Complete the statement.

$$\text{Area of triangle } CXD : \text{area of triangle } BXA = 2 : 1 \text{ [1]}$$

- 15 (a) Write 66 000 in standard form.

$$6.6 \times 10^4 \text{ [1]}$$

- (b) Work out $(3.7 \times 10^8) + (3.7 \times 10^7)$.

Give your answer in standard form.

$$(3.7 \times 10^8) + (0.37 \times 10^8)$$

$$4.07 \times 10^8$$

$$4.07 \times 10^8 \text{ [2]}$$